Mara Ulloa

PhD Candidate
U.S. National Science Foundation GRFP Fellow
Department of Computer Science
Northwestern University
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Research Interests

I specialize in using Human-Computer Interaction (HCI) methods to design and and evaluate human-centered Artificial Intelligence (AI).

I have worked with Apple on HCI + AI research; Microsoft on generative AI (genAI) for software engineering; Northwestern University on machine learning (ML) for health; The NSF National Center for Atmospheric Research on UX for data visualization software.

In my interdisciplinary doctoral research, I develop and apply participatory design methods to co-design ML across varied health contexts. More specifically, I focus on improving mental models of ML-driven interventions with and for laypersons.

Research interests/experience: Human-computer interaction; artificial intelligence/machine learning/Generative AI; design; UX; digital interventions; behavioral science; participatory design methods; software engineering; future of work; applied research.

Education

09/2021 - 06/2027 (expected) **Northwestern University (NU)**, Evanston, IL
Doctor of Philosophy (PhD), Computer Science, McCormick School of Engineering; GPA: 3.9/4.0
Research track: Human-Computer Interaction (HCI)
Committee:
Maia L. Jacobs, PhD (Chair & Advisor), NU
Jenna L. Butler, PhD, Microsoft
Michael Muller, PhD, Independent Researcher
Nabil Alshurafa, PhD, NU
Karan Ahuja, PhD, NU
Darius Tandon, PhD, NU

09/2021 - 06/2023 **Northwestern University**, Evanston, IL
Master of Science (MS), Computer Science, McCormick School of Engineering; GPA: 3.9/4.0

08/2018 - 05/2021 **Northeastern Illinois University (NEIU)**, Chicago, IL
Bachelor of Science (BS), Computer Science, College of Business and Technology; GPA: 3.8/4.0

2017 **Harold Washington College**, Chicago, IL
Associate in Arts (AA), General Studies; GPA: 3.8/4.0

Honors and Awards

2026 Keynote Speaker at NEIU CS Alumni Day
2026 David Hakken Participatory Design Grant
2025 Northwestern University CS - Graduate Women in Computing Ambassador Award
2025 Computing Research Association (CRA) - Grad Cohort
2023 - 2027 National Science Foundation (NSF) Graduate Research Fellowship (GRFP)

2023 - 2024 Center for Advancing Safety of Machine Intelligence (CASMI), Underwriters Laboratory (UL) - PhD Researcher Funding

2023 Computing Research Association (CRA) - WP Grad Cohort for Women

2022 Segal Design Research Cluster Fellowship

2022 NU Social Justice Mini Grant

2021 - 2022 NU CS Enhanced Funding Award

2021 NEIU CS Scholarship

2020 Abraham Lincoln Civic Engagement Award from the State of Illinois

2020 - ∞ NEIU Ronald E. McNair Scholar

2018 – 2020 The National TheDream.US Scholarship

2017 Phi Theta Kappa Honor Society

Publications

Peer-Reviewed Publications

Ulloa, M., Butler, J., Haniyur, S., Miller, C., Amos, B., Sarkar, A., Storey, M., "Product Manager Practices for Delegating Work to Generative AI: *'Accountability must not be delegated to non-human actors'* " *ACM/IEEE International Conference in Software Engineering (ICSE) December 13, 2025.*

Ulloa, M., Kamali, N., Fernandes, G., Soyemi, E., Beltzer M., Alshurafa, N., Jacobs, M., "Patient Expectations of ML-driven JITAI Support for Prenatal Stress Management" *ACM Transactions on Computing for Healthcare, Special Issue on Human Centered Computing in Healthcare (HEALTH), March 25, 2026.*

Miller, C., Choudhuri, R., **Ulloa M.**, Haniyur, D., DeLine, R., Storey, M., Murphy-Hill E., Bird C., Butler J. "'Maybe We Need Some More Examples:' Individual and Team Drivers of Developer GenAI Tool Use" *ACM/IEEE International Conference in Software Engineering (ICSE), October 16, 2025.* **Distinguished Paper Award**.

Cruz, S., Lu, C., **Ulloa M.**, Redding, A., Hester, J., Jacobs, M. "Perceptions of Wearable Health Tools Post-COVID 19 in Low-income Latine Communities" *JMIR mHealth and uHealth (JMU)*. USA, January 26, 2024.

Ulloa, M., Rothrock, B., Ahmad F., Jacobs, M. "Invisible Clinical Labor Driving the Successful Integration of AI in Healthcare." F*rontiers in Computer Science*. USA, December 1, 2022.

Workshop papers

Ulloa, M., Fernandes, G., Kamali, N., Soyemi E., Beltzer M., Kaveladze, B., Alshurafa, N.; Jacobs, M., "Co-Designing Patient-Facing Machine Learning for Prenatal Stress Reduction" *2024 CSCW: he 26th ACM Conference On Computer-Supported Cooperative Work And Social Computing (CSCW) - Supporting User Engagement in Testing, Auditing, and Contesting AI*. Minneapolis, MN, October 14 - 18, 2024.

ACM Queue

Butler, J., **Ulloa, M.**, Bird, C., Storey, M., "Climbing the Generative AI Mountain: A "Hitchhiker's" Guide for PMs" *ACM Queue*. May 18, 2026

Research Posters

Ulloa, M., Cherukuru, N., "User Experience (UX) of Advanced 3D Data Visualization Software" *U.S. National Science Foundation (NSF) National Center for Atmospheric Research (NCAR)*. Boulder, Colorado, USA August 7, 2024.

Ulloa, M., Kamali N.,Fernandes G., Soyemi e., Beltzer M., Kaveladze B., Alshurafa N., Jacobs, M., "Co-Designing Patient-Facing Machine Learning for Prenatal Stress Reduction" *The International Society for Research on Internet Interventions (ISRII)*. Limerick, Ireland, June 2 - 5, 2024.

Ulloa, M., & Jacobs, M., "Embedding Mental Health Support into Community CS Programs" *2023 CRA WP Grad Cohort Workshop for Women*. San Francisco, California, April 22, 2023.

Ulloa, M., and Jacobs, M., "Embedding Mental Health Support into Community CS Programs" *2022 Lambert Conference on the Future of Human-Computer Interaction + Design*. Evanston, Illinois, USA, October 24, 2022.

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Ulloa, M., Clark, S., Hatheway B., Weber J., Foust E., Wasson S., Ho Y., "Developing a Real-Time Weather Museum Touchscreen" *The National Center for Atmospheric Research: SIParCS and NCAR-Wyoming Presentations*. Boulder, Colorado, USA July 29, 2020.

Ulloa, M., Clark, S., Hatheway B., Weber J., Foust E., Wasson S., Ho Y., "Developing a Real-Time Weather Museum Touchscreen" *The National Center for Atmospheric Research Poster Presentations*. Boulder, Colorado, USA, July 30, 2020.

Ulloa, M., Clark, S., Hatheway B., Weber J., Foust E., Wasson S., Ho Y., "Developing a Real-Time Weather Museum Touchscreen" *UCLA National McNair Conference*. Los Angeles, California, USA, July 29-31, 2020.

Ulloa, M., Garcia J., Salem M., Gomez C., Adler R. F., Iacobelli F. , Trana R. "Creating a Large Cyberbullying Dataset from Multiple Sources to Improve the Accuracy of Cyberbullying Classifications" *2019 SACNAS The National Diversity in STEM Conference*. Honolulu, Hawaii, USA October 30 - November 3, 2019.

Garcia J., **Ulloa, M.**, Salem M., Gomez C., Adler R. F., Iacobelli F. , Trana R. " Cyberbullying classification: Semantic Analysis of Text in Shared Image Memes" *2019 SACNAS The National Diversity in STEM Conference*. Honolulu, Hawaii, USA October 30 - November 3, 2019.

Garcia J., Salem M., **Ulloa, M.**, Gomez C., Adler R. F., Iacobelli F., Trana R. "Creating a Large Cyberbullying Classification: Analysis of Text in Social Media Memes" *11th Annual SCSE Student Research Symposium*. Chicago, Illinois, USA, September 27, 2019.

Invited Talks

Ulloa, M. "ML-Driven System Design and the PhD Journey" *Roosevelt University Computer Science*. Chicago, IL, March 26, 2026.

Ulloa, M. "Keynote speech on my research and PhD experience" *Northeastern Illinois University Computer Science Alumni Day 2026*. Chicago, IL, March 25, 2026.

Ulloa, M., Fernandes, G., Kamali, N., Soyemi E., Beltzer M., Kaveladze, B., Alshurafa, N.; Jacobs, M.,"Co-Designing Patient-Facing Machine Learning for Prenatal Stress Reduction" *2024 CRA WP IDEALS*. Minneapolis, MN, April 13, 2024.

Ulloa, M. "GEO Community Practicum: Exploring the link between public scholarship & academic research" *Computer Science Department*. Evanston, IL, 2023.

Academic and Industry Experience

09/2021 - Present **Human-Computer Interaction (HCI) Researcher**
Northwestern University (NU), Evanston, IL

- Leading investigations by applying diverse HCI research methods and frameworks to investigate human-AI interaction
- Conducting multidisciplinary collaborations with principal investigators, research assistants, clinicians, health behavior scientists, and machine learning scientists
- Designing research plans, collecting and analyzing data, and distilling insights for both expert and lay audiences to inform the development of intelligent systems
- Identifying user needs, perspectives, goals, expectations, obstacles, and motivations to inform the design of intuitive and effective human-AI experiences
- Conducting multilingual research in both English and Spanish
- Authoring a range of academic artifacts, including research proposals, grant applications, manuscripts, workshop and research posters, and protocols for peer-reviewed dissemination
- Actively engagement in symposiums, research talks, literature reviews, writing workshops, courses, and seminars, staying at the forefront of HCI and AI

05/2026 - 08/2026 **HCI Researcher**
Apple, Cupertino, CA

- Lead multiple projects through the full research lifecycle: independently crafting and executing study formulation, participant recruitment, data collection and analysis (mixed methods).
- Reported results and actionable recommendations to team and leadership.
- Recruited for a 3.5 month PhD internship.

04/2025 - 07/2025 **Applied Research Scientist in Software Engineering**
Microsoft, Redmond, WA
Mentors: Dr. Jenna Butler, Dr. Margaret-Anne Storey, Dr. Advait Sarkar, Sankeerti Haniyur, Barrett Amos

- Selected from a pool of 300+ PhD intern applicants
- Lead the design and execution of a three-month, mixed-methods study to investigate the impact of generative AI in software engineering, specifically focusing on the understudied product manager role.
- Conducted exploratory research, gathering 885 quantitative and qualitative user surveys, telemetry data, and 15 in-depth interviews from participants across the globe, complemented by thorough secondary research including literature reviews and internal company data analysis.
- Leveraging Python scripts for quantitative analysis and applied thematic analysis, diagramming, and collaborative workshops for qualitative insights.
- Presented research progress and key findings to diverse stakeholders, including Corporate Vice Presidents, Partner Managers, Product Managers, and our research team.
- Disseminated research through both academic artifacts (e.g., papers) and tailored communications for non-academic audiences, such as marketing, developers, and business + AI groups.
- Delivered clear, actionable next steps, with proposed interventions designed to extend impact (beyond product managers) across product and other user groups.
- This work forms part of Microsoft's 2026 New Future of Work Report

05/2024 - 08/2024 **User Experience (UX) Researcher of Advanced 3D Data Visualization Software**
U.S. National Science Foundation National Center for Atmospheric Research (NSF NCAR), Boulder, CO
Mentor: Dr. Nihanth Cherukuru

- Led a three-month exploratory UX research study, employing both primary (in-depth user interviews) and secondary research (literature reviews) to understand user needs within advanced data visualization software.
- Applied advanced UX methodologies, specifically Jobs-to-be-Done (JTBD) theory, to uncover end-user needs, preferences, and pain points related to complex data visualization software tools.
- Conducted over 10 user interviews, performing detailed data transcription and rigorous analysis to deliver actionable insights and recommendations to both the project team and broader organization.
- Leveraged strong analytical skills to meticulously interpret qualitative research data, identify critical user trends, and synthesize findings into actionable conclusions for software development and product improvement.
- Presented comprehensive research findings through both oral presentations and poster sessions to the NSF NCAR community, advocating for user-centered design and contributing to community knowledge.
- Actively participated in weekly professional development workshops, enhancing skills in science communication, leadership, proposal writing, and data management and visualization.

05/2020 - 08/2020 **Software Engineer Intern**
U.S. National Science Foundation National Center for Atmospheric Research (NSF NCAR), Boulder, CO
Mentor: Sharon Clark

- Designed, developed and deployed a new touchscreen interface in Java for real-time weather data visualization, utilizing prototypes to create a user-friendly and educational kiosk application installed at NCAR's Mesa Lab and Wyoming Supercomputing Center.
- Conducted virtual user interviews and facilitated communication with stakeholders to gather and define application requirements.
- Led team meetings, facilitating discussions, and meticulously transcribing detailed notes of plans, decisions, and options to ensure clear communication and progress.
- Participated in ten weekly science communication workshops, developing enhanced skills in presenting technical information and fostering collaboration through team-building discussions.
- Presented project outcomes and technical insights through both poster and oral presentations to the scientific communities at NCAR and the National Oceanic and Atmospheric Administration (NOAA).

08/2019 - 05/2021 **Computer Science Society President**
Northeastern Illinois University (NEIU), Chicago, IL

- Attended meetings with faculty advisor and other officers on implementation of student-led biweekly workshops
- Led team responsible for maintaining the society's website
- Designed informative recruitment materials for workshops
- Coordinated workshops provided by a team of officers on tools such as GitHub and JavaScript fundamentals

05/2019 - 11/2019 **Undergraduate Research Assistant**
Northeastern Illinois University (NEIU), Chicago, IL
Mentors: Dr. Rachel Adler, Dr. Rachel Trana, Dr. Francisco Iacobelli

- Applied text classification algorithms that identified text as harmful or non-harmful with 87% accuracy with the goal of improving the accuracy of cyberbullying algorithmic classifications
- Independently leaned Python and writing python scripts to retrieve and classify new socially relevant text images from existing datasets
- Engaged in weekly career development workshops
- Researched various subjects that were related to the project and presented information to mentors and team
- Presented findings at SACNAS National Conference and NEIU's Student Center for Science Engagement Symposium

Teaching Experience

01/2026 - Present **Instructor of Human Computer Interaction Studio**
Northwestern University (NU), Evanston, IL

- Instructing (twice a week) to introduce 50 undergraduate and masters students to foundational HCI concepts and techniques + guiding hands-on interaction design and user testing
- Co-instructing the course in collaboration with professors, teaching assistants, and peer mentors through both synchronous and asynchronous communication
- Conducting studio and office hour sessions for students to gain feedback on their quarter-long design projects

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03/2022 - 03/2024 **Coding Instructor and Research Collaborator**
We All Code (WAC), Chicago, IL

- Delivered engaging six-hour workshops on programming fundamentals to 15 - 30 youth (ages 7-18), utilizing both in-person and virtual modalities.
- Collaborated with the organization's founder to strategize student engagement and continuously enhance learning experiences, leading to improved outcomes.
- Designed compelling recruitment materials and newsletters, drafted grant proposals, and managed external communications to support program growth.
- Designed classroom environments conducive to effective learning, warmly welcomed students and families, and effectively managed a team of mentors.
- Maintained and managed specialized classroom equipment, including robotics kits and laptops, ensuring seamless operation for our workshops.
- Distilled insights from team retrospectives to iteratively refine course content and instructional methods, ensuring continuous improvement in learning delivery.

01/2025 **Instructor - Computing Everywhere: Tech-Facilitated Abuse & Trauma-Informed Design**
Northwestern University (NU), Evanston, IL

- Crafted and submitted a proposal for this course to NU's Human-Computer Interaction + Design (HCI+D) Computing Everywhere program
- In collaboration with my colleague, Connie Chau, developed & taught first-ever curriculum on tech-facilitated abuse, digital safety, and trauma-informed design
- Adapted the class for non-computing an interdisciplinary undergraduates
- Consulted in a formal setting with other PhD instructors to discuss course logistics and share pedagogical approaches prior to teaching

08/2019 - 05/2021 **Computer Science Peer Leader**
Northeastern Illinois University (NEIU), Chicago, IL

- Facilitated weekly workshops for over 240 undergraduate students, covering foundational Programming I and II concepts in both virtual and in-person settings.
- Collaborated with professors and other peer leaders to optimize workshop content and delivery, significantly enhancing the learning experience for students.
- Mentored students by guiding them to essential campus resources and external opportunities, fostering their academic and professional development.
- Actively engaged in comprehensive Peer Leader training, developing advanced skills in instructional techniques, communication, and leadership.

Skills

Research Methods	Affinity diagramming, A/B Testing, Co-design, Contextual Inquiry, Design sessions, Field Studies, Focus Groups, Heuristic evaluation, Jobs-to-be-done, Journey Mapping, Mixed Methods Research, Participatory Design, Qualitative analysis, Hi/Low-fidelity Prototyping, Storyboarding, Surveys, Thematic Analysis, Usability Testing, User Interviews, Wizard-of-Oz Prototyping
UX Tools	Figma, Canva, NinjaMock, Miro, Mural, Sketch
Research Tools	Qualtrics, REDCap, Microsoft Forms, Otter.ai, Google Analytics, Reference Managers
Programming	Java, Python, HTML, CSS, SQL
Version Control	GitHub
Languages	English, Spanish

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Service

Conference Service

Reviewer — ACM CHI Conference on Human Factors in Computing Systems 2026
Reviewer — ACM IMWUT: Interactive, Mobile, Wearable & Ubiquitous Tech. 2025
Reviewer — ACM CHI: Conference on Human Factors in Computing Systems 2024
Reviewer — ACM TOCHI: Transactions on Computer-Human Interaction 2024
Reviewer — ACM CHI: Conference on Human Factors in Computing Systems 2023
JEDI (Justice, Equity, Diversity, & Inclusion) **Ambassador** — UbiComp/ISWC 2023
Volunteer — ACM IDC: Interaction Design and Children 2023

University Service

NU The Graduate School Office of Diversity and Inclusion (ODI) Diversity Peer Mentor
NU CS Diversity, Equity and Inclusion Minute — Presenter
NU Future of Health and Computing Collective (FHCC) — Student Committee
NU CS Graduate Women in Computing (GWiC) — Board Member
NU CS PhD Student Visit Day — Planning Committee
NU SROP Pathways to the Professoriate & Navigating Graduate School — Panelist
NU CS PhD Student Visit Day — Planning Committee
NU Computer Science Social Initiative (CSSI) — Board Member
NEIU Alumni Panel — Panelist
NEIU SCSE, TRIO, and McNair Graduate School Panel — Panelist

Community Service

We All Code
Pancreatic Cancer Action Network
Chicago Canine Rescue
Cornerstone Autism Services
Friendship Center
NEIU Food Pantry
Leukemia & Lymphoma Society
Lurie Children's Hospital

Selected Press

2026 **Patients Demand More from AI Health Tools than Just Answers**
2024 **PhD Student Spotlight: Mara Ulloa**
2024 **The AI Will See You Now**
2023 **Patient-Focused AI System Seeks to Reduce Stress during Pregnancy**
2023 **Building Mentoring Relationships and Peer Networks at CRA-WP Grad Cohort for Women**
2022 **Supporting Mental and Emotional Health Within Computer Science Education**
2022 **Mara Ulloa Awarded NSF Graduate Research Fellowship**
2020 **Computer Science major Mara Ulloa named Lincoln Academy Student Laureate**

Affiliations

Center for Advancing Safety of Machine Intelligence (CASMI)
Computer Science Society
McNair Scholars Program
Phi Theta Kappa Honor Society
Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS)
The Lincoln Academy of Illinois

Student Mentoring

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Grace Yang, BS CS (NU, class of 2026; Now Software Engineer at Snowflake)
Niharika G. Menon, MS CS (NU, class of 2024; Now Sr. Product Analyst at Avant)
Elizabeth Soyemi, BA Public Health (Brown University, class of 2026)
Kevin Baez, BS CS, MS BA (NEIU, class of 2027)
Cam Portland, BS CS & Cognitive Science, (NU, class of 2028)
Vanessa Ikirezi, BS CS (NU, class of 2028)
Mahnum Somji, BS CS (NU, class of 2028)

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